Root Cause Troubleshooting Guide

MVTec AD Industrial Defect Inspection Knowledge Base

For RAG-powered troubleshooting, safe root-cause suggestions, and operator decision support

Document type	Synthetic QA / SOP guide for Retrieval-Augmented Generation
Project	AI Defect Detection and Root-Cause Assistant
Dataset context	MVTec AD selected categories: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut
Main use	Retrieve likely causes, inspection checkpoints, and next troubleshooting actions after computer vision detects an anomaly
Important limitation	This guide provides possible causes and recommended checks. It must not be used to claim a guaranteed root cause from image data alone.
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Use this document as one of the RAG source files in the project knowledge base. The language is intentionally structured with repeated product names, defect terms, action terms, and cause terms so that chunking and semantic retrieval work reliably.

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Note: Page numbers are intentionally omitted because this PDF is designed mainly for RAG ingestion and document retrieval. Section headings should be used as chunk anchors.

1. Purpose and Scope

This Root Cause Troubleshooting Guide supports the AI Defect Detection and Root-Cause Assistant project. The guide explains possible root causes, inspection checkpoints, and recommended troubleshooting steps for visible product anomalies in selected MVTec AD categories.

The guide is designed for a system where a computer vision model first detects an anomaly from a product image. A RAG retriever then searches this document to provide evidence-based troubleshooting suggestions. The final answer may be written by an LLM, but the LLM must stay grounded in this guide and in the vision model output.

Covered MVTec AD categories

- Bottle
- Cable
- Screw
- Transistor
- Tile
- Toothbrush
- Zipper
- Metal nut

> This guide does not replace real factory engineering analysis. It is a student-project knowledge base and should be used as a realistic simulated industrial document. Image-only inspection cannot confirm every root cause. Many true causes require machine logs, material records, operator notes, sensor readings, or destructive testing.

2. How the AI Assistant Must Use This Guide

The AI assistant should use this document after the vision model detects or suspects an anomaly. The assistant should retrieve the relevant category section and then use the visual signs, confidence score, severity level, and retrieved evidence to write a safe troubleshooting response.

Required input signals

Signal	Meaning	How it affects troubleshooting
Product category	The object being inspected, such as bottle, cable, screw, transistor, tile, toothbrush, zipper, or metal_nut.	Selects the correct troubleshooting section.
Anomaly status	Normal, suspected anomaly, or confirmed anomaly.	Determines whether troubleshooting is needed.
Anomaly location	Where the anomaly appears, such as edge, surface, thread, pin area, tooth area, or connector area.	Narrows likely process causes.
Model confidence	How confident the vision model is.	Low confidence requires human review and careful wording.
Severity level	Minor, major, critical, or human review required.	Controls action urgency.
OCR or metadata	Batch code, label text, serial number, or production notes if available.	May support batch-level investigation.

Required assistant behavior

- Use the phrase 'possible root cause' or 'likely inspection checkpoint' unless strong external evidence is available.
- Do not state that a root cause is definite based only on an image.
- If the retrieved evidence does not match the detected category, say that the evidence is insufficient.
- If image quality is poor or confidence is low, recommend human review before final action.

- For critical anomalies, recommend quarantine or supervisor review even when the model confidence is high.

3. Root-Cause Language Policy

Root-cause language must be careful. A product image can show symptoms, but the true process cause may need additional data. The assistant must distinguish visual evidence from engineering proof.

Unsafe wording	Preferred safe wording
The crack was caused by high machine pressure.	The crack may be associated with high pressure, impact, or cooling stress. Check molding pressure and handling records.
The cable failed because insulation was dirty.	The visual anomaly may indicate contamination or insulation damage. Inspect material cleanliness and extrusion setup.
The screw defect is due to tool wear.	The thread anomaly is consistent with possible tool wear, poor alignment, or feed issue. Check tooling condition and alignment.
The transistor is defective because soldering failed.	The visible issue may be related to component placement, bent leads, missing part, or soldering/assembly handling. Inspect assembly records.

Root-cause confidence levels

Level	Use when	Allowed wording
Low	Only visual anomaly exists; no process data or metadata is available.	Possible cause; recommended checkpoint; needs human review.
Medium	Visual anomaly plus matching SOP evidence exists.	Likely cause category; inspect specific process step.
High	Visual anomaly plus matching SOP evidence plus batch/machine/inspection data exists.	Strongly indicated cause; confirm with engineering review.

4. Standard Troubleshooting Workflow

The following workflow should be followed by operators, QA engineers, or the AI assistant after a defect or anomaly is detected.

1. Verify image quality: confirm the product is fully visible, focused, well lit, and not blocked by shadows or reflections.
2. Confirm product category: bottle, cable, screw, transistor, tile, toothbrush, zipper, or metal nut.
3. Confirm anomaly status and location: surface, edge, joint, connector, pin, thread, tooth, bristle, or internal region.
4. Check severity guide: decide whether the issue is minor, major, critical, or needs human review.
5. Retrieve the category-specific troubleshooting section from this document.
6. List possible root-cause families, not one guaranteed root cause.
7. Recommend immediate containment: accept, rework, reject, quarantine, retake image, or human review.
8. Recommend investigation checkpoints: machine setting, tool wear, material condition, handling, cleaning, alignment, or packaging.
9. Record the result in the inspection report with evidence and limitations.

Immediate containment rules

Detected condition	Immediate containment	Reason
Normal image and high confidence	Accept or continue normal process	No visible anomaly detected.
Minor cosmetic anomaly	Rework or monitor depending on product category	May not affect function but should be tracked.
Major visible damage	Reject or remove from line	Product may fail quality requirements.

Detected condition	Immediate containment	Reason
Critical structural or safety issue	Quarantine batch and escalate	Possible safety, function, or customer-impact risk.
Low confidence or poor image	Retake image or human review	AI result is not reliable enough.

5. General Process Cause Taxonomy

Most visible industrial anomalies can be grouped into cause families. These cause families help the assistant explain the issue clearly and guide the operator toward practical checks.

Cause family	Typical visual signs	Example checkpoints
Material issue	Color variation, contamination, surface inclusion, brittleness, stain, foreign particles.	Supplier lot, raw material inspection, storage condition, moisture, contamination control.
Machine setting issue	Repeated pattern, deformation, cracks, uneven surface, incomplete formation.	Pressure, temperature, speed, feed rate, cooling time, extrusion or molding parameters.
Tooling or fixture issue	Scratches, dents, thread damage, repeated marks, misalignment.	Tool wear, die condition, fixture alignment, cutting tool sharpness, clamp pressure.
Handling or transport issue	Random scratches, broken edges, impact marks, crushed areas.	Operator handling, conveyor contact, bin storage, packaging, drop events.
Assembly issue	Missing component, wrong orientation, bent pin, loose part, misplacement.	Pick-and-place accuracy, feeder setup, assembly jig, soldering or fastening process.
Cleaning or contamination issue	Dust, oil, stain, residue, speckled anomaly.	Cleaning cycle, compressed air, oil leak, environmental dust, gloves, workspace hygiene.
Inspection setup issue	False anomaly due to glare, shadow, blur, partial image.	Camera focus, lighting, lens cleanliness, angle, background, image resolution.

6. Evidence Required Before Suggesting a Cause

The assistant should not suggest root causes without matching evidence. Evidence can come from image analysis, retrieved documents, OCR, metadata, or human notes.

Evidence type	Examples	Usage
Visual evidence	Anomaly location, size, shape, texture, color, defect mask, bounding box.	Required for all defect explanations.
Document evidence	QA manual rule, SOP action, severity guide, troubleshooting note.	Required for RAG-grounded explanation.
Metadata evidence	Category, batch ID, machine ID, production line, inspection time.	Useful for batch-level root-cause investigation.
Process data	Temperature, pressure, feed speed, torque, vibration, tool age.	Needed for higher confidence cause claims.
Human review evidence	Inspector notes, supervisor decision, lab test.	Needed for final confirmation.

Minimum evidence rule

Minimum evidence for a RAG answer: product category + defect/anomaly description + one retrieved section from a relevant document. If the system cannot retrieve a relevant section, it should answer: 'No matching troubleshooting evidence was found in the current knowledge base; human review is required.'

7. Category-Specific Troubleshooting Guides

Each category section below is intended to be retrieved by RAG when the assistant needs product-specific troubleshooting evidence. The terms are repeated intentionally to improve retrieval for natural-language questions.

7.1 Bottle Root-Cause Troubleshooting

Product context: Bottle inspection focuses on the container body, rim, base, side wall, opening, and external surface. Common risks include cracks, contamination, deformation, broken edges, and foreign material.

Common visual indicators

- Crack or fracture line on body, rim, neck, or base
- Dented or deformed wall
- Contamination mark, stain, or foreign particle
- Broken or chipped edge
- Unusual shape variation compared with good bottles

Possible root-cause families

Possible cause family	Explanation
Molding or forming parameter issue	High or unstable pressure, temperature variation, poor cooling, or incorrect cycle time may create cracks, deformation, or weak areas.
Material or contamination issue	Impure resin, glass impurity, moisture, or mixed raw material may create spots, inclusions, weak surfaces, or color variation.
Handling or impact issue	Drops, conveyor collision, bin contact, or rough packaging may create edge chips, cracks, and impact marks.
Inspection setup issue	Strong reflection, transparent material glare, or poor contrast can look like a defect even when the bottle is normal.

Recommended troubleshooting checks

- Check mold temperature, pressure, and cooling records for the inspected batch.
- Inspect conveyor contact points and packaging bins for impact sources.
- Verify raw material batch and storage condition.
- Retake image using diffuse lighting if glare hides transparent surfaces.
- If rim or sealing area is damaged, treat as major or critical and quarantine the item.

Immediate actions

- Reject bottles with cracks, holes, broken rim, or sealing-area damage.
- Quarantine repeated crack defects from the same batch.
- Rework is allowed only for removable surface contamination if the product standard allows cleaning.
- Send low-confidence transparent-bottle anomalies to human review.

RAG retrieval keywords: bottle crack, bottle contamination, bottle broken, bottle deformation, bottle rim defect, bottle root cause, bottle troubleshooting, molding issue, cooling issue, handling impact

7.2 Cable Root-Cause Troubleshooting

Product context: Cable inspection focuses on insulation surface, wire continuity area, connectors, bending regions, exposed conductor risk, and contamination marks. Cable defects can affect electrical safety and reliability.

Common visual indicators

- Cut, nick, puncture, or exposed conductor
- Burn mark, dark stain, or contamination
- Bent or twisted cable geometry
- Damaged connector or missing cable segment
- Surface bubble, crack, or insulation irregularity

Possible root-cause families

Possible cause family	Explanation
Insulation extrusion issue	Incorrect extrusion temperature, poor material flow, or unstable line speed may create bubbles, cracks, thickness variation, or weak insulation.
Tooling or stripping issue	Sharp tools, incorrect blade depth, or worn cutters can nick insulation or expose conductor.
Handling and routing issue	Over-bending, tight fixture, conveyor pinch, or packaging pressure may deform or damage cable sections.
Contamination issue	Oil, dust, resin residue, or environmental particles can appear as stains or surface spots.

Recommended troubleshooting checks

- Inspect extrusion temperature, pressure, and line speed logs.
- Check stripping blade depth and cutter wear.
- Check bending radius and fixture contact points.
- Verify connector assembly alignment and crimping condition.
- Use electrical test results if available before final acceptance.

Immediate actions

- Reject exposed conductor, deep cuts, burned insulation, or damaged connector conditions.
- Human review is required for ambiguous stains near conductor areas.
- Quarantine batch if repeated insulation damage appears in the same cable location.
- Retake image if cable is twisted or hidden by shadows.

RAG retrieval keywords: cable insulation defect, exposed conductor, cable cut, cable contamination, cable connector damage, extrusion issue, blade depth, crimping issue, cable root cause

7.3 Screw Root-Cause Troubleshooting

Product context: Screw inspection focuses on thread condition, head shape, shaft straightness, tip quality, corrosion, surface scratches, and missing or deformed thread regions.

Common visual indicators

- Damaged or missing thread
- Bent screw shaft
- Deformed head slot or drive recess
- Rust, discoloration, or contamination
- Surface scratch, dent, burr, or broken tip

Possible root-cause families

Possible cause family	Explanation
Thread rolling or cutting issue	Worn dies, poor alignment, incorrect feed, or low lubrication can create damaged thread, burrs, or incomplete thread formation.
Tool wear issue	Worn heading tools or drive-forming tools can create deformed heads, poor recess shape, or repeated marks.
Material hardness issue	Incorrect hardness, brittle material, or inconsistent wire stock can lead to cracks, broken tip, or poor thread quality.
Storage or corrosion issue	Humidity, poor coating, or improper storage may cause rust or discoloration.

Recommended troubleshooting checks

- Inspect thread rolling die condition and alignment.
- Check lubrication and feed speed settings.

- Check wire material certificate and hardness result.
- Inspect plating or coating process records.
- Use gauge check for thread acceptability if available.

Immediate actions

- Reject screws with missing thread, broken tip, severe rust, or damaged drive recess.
- Rework may be possible only for removable surface contamination, not structural thread damage.
- Escalate repeated same-location thread damage to tooling maintenance.
- Human review is required if image does not show the full screw profile.

RAG retrieval keywords: screw thread defect, damaged thread, screw head deformation, screw rust, screw burr, thread rolling issue, tool wear, screw root cause

7.4 Transistor Root-Cause Troubleshooting

Product context: Transistor inspection focuses on the component body, printed markings, lead pins, package edge, component orientation, pin straightness, missing or bent parts, and assembly integrity.

Common visual indicators

- Bent, broken, missing, or misaligned lead pin
- Cracked or chipped transistor body
- Wrong orientation or shifted placement
- Missing component region or damaged package corner
- Unreadable or incorrect marking

Possible root-cause families

Possible cause family	Explanation
Pick-and-place or assembly issue	Misalignment, incorrect feeder setup, poor vacuum pickup, or placement force can bend leads or shift components.
Handling issue	Manual handling, tray contact, or packaging pressure can bend pins or chip package corners.
Soldering or thermal stress issue	Excess heat, poor temperature profile, or rework damage may affect body or pin integrity.
Supplier or packaging issue	Incoming components may already have bent leads, wrong marking, or damaged package if packaging is poor.

Recommended troubleshooting checks

- Inspect feeder alignment and pick nozzle condition.
- Check component tray orientation and incoming inspection records.
- Check soldering temperature profile and rework history if PCB assembly is involved.
- Verify marking text with OCR if marking quality matters.
- Use electrical test results for final confirmation.

Immediate actions

- Reject missing lead, broken package, severe bent lead, wrong orientation, or unreadable critical marking.
- Human review required for minor marking defects or low-confidence pin anomalies.
- Quarantine if repeated lead bending appears for the same supplier lot or feeder lane.
- Escalate to process engineer if component placement errors repeat.

RAG retrieval keywords: transistor bent pin, transistor missing lead, transistor package crack, transistor orientation, component placement, pick and place issue, soldering stress, transistor root cause

7.5 Tile Root-Cause Troubleshooting

Product context: Tile inspection focuses on surface texture, edge quality, cracks, chips, stains, pattern uniformity, glazing quality, and flatness. Tile defects can be cosmetic or structural.

Common visual indicators

- Surface crack, corner chip, or edge break
- Stain, spot, discoloration, or contamination
- Uneven texture, glaze defect, or scratch
- Pattern mismatch or surface inclusion
- Warping, deformation, or visible fracture line

Possible root-cause families

Possible cause family	Explanation
Firing or cooling issue	Thermal stress, uneven firing temperature, or rapid cooling can create cracks, warping, or surface variation.
Raw material or mixing issue	Impurities, poor clay mix, moisture variation, or granule contamination can create spots, inclusions, or weak areas.
Glazing or coating issue	Poor glaze viscosity, uneven spraying, contamination, or kiln atmosphere can create stains and color variation.
Handling or cutting issue	Conveyor impact, stacking pressure, or cutting damage can create chips and broken edges.

Recommended troubleshooting checks

- Check kiln temperature profile and cooling curve.
- Verify raw material moisture and mixing batch.
- Inspect glazing nozzle, viscosity, and contamination controls.
- Check packaging, stacking, and conveyor impact points.
- For edge chips, check cutting and handling process.

Immediate actions

- Reject cracked tiles, broken corners, severe warping, or large visible stains.
- Minor cosmetic marks may be downgraded or reworked only if customer standard allows.
- Quarantine repeated same-pattern surface defects from the same kiln batch.
- Retake image if tile reflection or texture pattern causes uncertain detection.

RAG retrieval keywords: tile crack, tile chip, tile stain, tile glaze defect, tile surface defect, kiln temperature, cooling stress, raw material contamination, tile root cause

7.6 Toothbrush Root-Cause Troubleshooting

Product context: Toothbrush inspection focuses on bristle condition, head integrity, handle surface, molded shape, color marks, contamination, and missing or deformed bristle groups.

Common visual indicators

- Missing, bent, uneven, or contaminated bristles
- Cracked toothbrush head or handle
- Flash, molding mark, dent, or rough edge
- Color inconsistency or stain
- Foreign particle near bristle area

Possible root-cause families

Possible cause family	Explanation
Bristle insertion issue	Incorrect tufting force, poor filament feeding, or worn tufting tool can create missing, uneven, or loose bristles.
Injection molding issue	Incorrect mold temperature, pressure, or cooling can create sink marks, flash, cracks, or shape deformation.
Material or colorant issue	Contaminated resin, wrong colorant ratio, or moisture can create stains, color variation, or weak molded areas.
Handling or hygiene issue	Open storage, poor cleaning, or packaging contamination can introduce particles in bristle area.

Recommended troubleshooting checks

- Check tufting machine settings and filament feed path.
- Inspect mold temperature, injection pressure, and cooling cycle.
- Verify resin and colorant batch records.
- Inspect cleaning and packaging environment for contamination.
- Check pull-strength tests for bristle retention if available.

Immediate actions

- Reject missing bristles, foreign material in bristle area, cracked head, or unsafe sharp edge.
- Quarantine if repeated bristle defects appear from the same tufting machine.
- Rework may be possible only for approved cleaning of external contamination, not missing bristles.
- Human review required for minor color variation if cosmetic standard is unclear.

RAG retrieval keywords: toothbrush bristle defect, missing bristle, uneven bristles, toothbrush contamination, injection molding issue, tufting issue, toothbrush root cause

7.7 Zipper Root-Cause Troubleshooting

Product context: Zipper inspection focuses on teeth alignment, slider condition, tape quality, stitching, missing teeth, broken teeth, deformation, contamination, and smooth closing function.

Common visual indicators

- Missing, broken, or misaligned zipper teeth
- Damaged slider or pull tab
- Torn or contaminated tape
- Uneven stitching or loose thread
- Bent, jammed, or separated teeth region

Possible root-cause families

Possible cause family	Explanation
Teeth molding or forming issue	Poor tooth formation, worn mold, incorrect pressure, or material variation can create weak or missing teeth.
Assembly alignment issue	Tape misalignment, incorrect slider installation, or poor stitching can lead to jammed or separated teeth.
Handling or mechanical stress	Pulling force, bending, packaging compression, or sharp contact can break teeth or deform slider.
Contamination issue	Dust, oil, thread residue, or foreign material can affect slider movement or visual cleanliness.

Recommended troubleshooting checks

- Inspect teeth forming mold and material feed.
- Check slider installation, tape alignment, and stitching process.
- Perform open-close functional test if available.
- Check packaging pressure and bending during transport.
- Inspect for foreign material around tooth rows.

Immediate actions

- Reject missing teeth, broken slider, severe tape tear, or functional jam.
- Human review required if only cosmetic dirt is visible and function is unknown.
- Quarantine repeated tooth alignment defects from same machine setup.
- Rework may be possible only for removable contamination or loose thread trimming if standard allows.

RAG retrieval keywords: zipper missing teeth, zipper broken teeth, zipper slider defect, zipper tape tear, zipper stitching, teeth alignment issue, zipper root cause

7.8 Metal Nut Root-Cause Troubleshooting

Product context: Metal nut inspection focuses on inner thread, outer surface, edges, flatness, corrosion, machining marks, dents, cracks, contamination, and dimensional appearance.

Common visual indicators

- Damaged or missing internal thread
- Rust, corrosion, oil stain, or contamination
- Crack, chip, burr, or sharp edge
- Deformed hex shape or dent
- Surface scratch, machining mark, or discoloration

Possible root-cause families

Possible cause family	Explanation
Machining or tapping issue	Worn tap, poor alignment, incorrect feed, or chip buildup can create damaged internal threads and burrs.
Tooling or fixture issue	Clamp pressure, worn fixture, or misalignment can deform surfaces or create repeated marks.
Material or heat treatment issue	Incorrect hardness, brittleness, or inconsistent stock may cause cracking, chipping, or poor thread formation.
Cleaning, coating, or storage issue	Oil residue, plating defect, moisture, or poor storage can create stains, rust, or contamination.

Recommended troubleshooting checks

- Inspect tapping tool condition and alignment.
- Check coolant/lubrication and chip removal process.
- Verify material hardness and heat treatment records.
- Inspect plating, cleaning, and storage conditions.
- Use thread gauge and dimensional checks for final acceptance.

Immediate actions

- Reject nuts with damaged internal thread, crack, severe corrosion, or deformation.
- Rework may be possible for removable oil or light contamination only if dimensional quality is acceptable.
- Escalate repeated thread damage to tooling maintenance.

- Human review required if inner thread is not clearly visible in the image.

RAG retrieval keywords: metal nut thread damage, nut corrosion, metal nut burr, nut deformation, tapping issue, tool wear, plating defect, metal nut root cause

8. Cross-Category Cause Matrix

This matrix helps the assistant map a visible anomaly to likely investigation checkpoints across categories. It should not be used to guarantee a single root cause.

Visual symptom	Likely cause families	Applies strongly to	Recommended checkpoint
Crack or fracture	Thermal stress, impact, material brittleness, pressure issue	Bottle, tile, transistor, toothbrush, metal nut	Check temperature/cooling, handling, material lot, hardness, molding or firing records.
Scratch or surface mark	Handling contact, tooling contact, conveyor impact, contamination	Bottle, screw, tile, metal nut, toothbrush	Check conveyor, storage bins, fixtures, tool surfaces, packaging.
Missing part or missing feature	Assembly error, forming issue, machine feed issue	Transistor, zipper, screw, toothbrush	Check feeder, tool setup, insertion process, assembly jig, operator station.
Contamination, stain, residue	Cleaning failure, oil leak, foreign particle, storage issue	Bottle, cable, tile, toothbrush, zipper, metal nut	Check cleaning cycle, oil sources, storage condition, packaging environment.
Misalignment or deformation	Fixture issue, feed alignment, assembly placement, mechanical stress	Cable, screw, transistor, zipper, metal nut	Check alignment, fixture clamping, feeder path, handling pressure.
Corrosion or discoloration	Moisture, coating problem, chemical exposure, storage issue	Screw, metal nut, cable	Check humidity, coating/plating, storage duration, material protection.

9. Human Review and Escalation Rules

The AI assistant should ask for human review or escalation when the visual evidence is not enough, the risk is high, or the retrieved evidence is weak.

Condition	Required response	Reason
Vision confidence below 70 percent	Human review required	The AI prediction may be unreliable.
Image is blurry, dark, overexposed, cropped, or product is partially hidden	Retake image before root-cause answer	Defect signs may be false or missed.
Critical safety/function issue	Quarantine and supervisor review	Risk may affect user safety or product function.
RAG evidence does not match product category	Do not provide specific root cause	Retrieved document may be irrelevant.
Repeated same defect in same category or batch	Escalate process investigation	May indicate systematic process fault.
Only image available and no process data	Use possible-cause language	Image alone cannot prove process cause.

10. RAG Retrieval Keywords and Chunking Notes

This document should be chunked by section heading and product category. Each chunk should keep product name, visual symptom, possible cause, and action together. Do not split a category table from its category heading if possible.

Recommended chunking strategy

- Chunk by main section first, then by product category.

- Use chunk size around 500 to 900 tokens with overlap around 80 to 120 tokens.
- Preserve headings such as 'Bottle Root-Cause Troubleshooting' and 'Cable Root-Cause Troubleshooting'.
- Store metadata fields: document_name, section_title, product_category, document_type, version.
- For category chunks, metadata product_category should exactly match: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut.

Example metadata

```
document_name: Root_Cause_Troubleshooting_Guide.pdf section_title: Screw Root-Cause
Troubleshooting product_category: screw document_type: troubleshooting_guide version: 1.0
```

Important retrieval terms

Product	Useful retrieval terms
Bottle	bottle crack, bottle contamination, bottle broken, bottle deformation, bottle rim defect, bottle root cause, bottle troubleshooting, molding issue, cooling issue, handling impact
Cable	cable insulation defect, exposed conductor, cable cut, cable contamination, cable connector damage, extrusion issue, blade depth, crimping issue, cable root cause
Screw	screw thread defect, damaged thread, screw head deformation, screw rust, screw burr, thread rolling issue, tool wear, screw root cause
Transistor	transistor bent pin, transistor missing lead, transistor package crack, transistor orientation, component placement, pick and place issue, soldering stress, transistor root cause
Tile	tile crack, tile chip, tile stain, tile glaze defect, tile surface defect, kiln temperature, cooling stress, raw material contamination, tile root cause
Toothbrush	toothbrush bristle defect, missing bristle, uneven bristles, toothbrush contamination, injection molding issue, tufting issue, toothbrush root cause
Zipper	zipper missing teeth, zipper broken teeth, zipper slider defect, zipper tape tear, zipper stitching, teeth alignment issue, zipper root cause
Metal Nut	metal nut thread damage, nut corrosion, metal nut burr, nut deformation, tapping issue, tool wear, plating defect, metal nut root cause

11. Sample RAG Questions and Expected Evidence

The following questions can be used to test whether the RAG retriever returns the correct section from this guide.

Sample user question	Expected retrieved evidence
The model found a crack on a bottle rim. What could be the cause?	Bottle troubleshooting section; molding/forming issue, cooling issue, handling impact, reject rim damage.
A cable image shows a dark stain near the insulation. What should we check?	Cable troubleshooting section; contamination issue, extrusion issue, cleaning/oil/dust checks.
The screw thread looks damaged. What is the likely process checkpoint?	Screw troubleshooting section; thread rolling/cutting issue, die wear, alignment, lubrication.
The transistor has a bent lead pin. What is the possible root cause?	Transistor troubleshooting section; pick-and-place, feeder alignment, handling, packaging pressure.
A tile has a crack and surface stain. What should the operator investigate?	Tile troubleshooting section; firing/cooling issue, raw material, glazing, handling.
The toothbrush has missing bristles. What action should be recommended?	Toothbrush troubleshooting section; tufting issue, reject missing bristles, check tufting machine.
The zipper has missing teeth. What could cause it?	Zipper troubleshooting section; teeth forming issue, assembly alignment, handling stress.
The metal nut has damaged internal thread. What should we check?	Metal nut troubleshooting section; tapping issue, tool wear, alignment, thread gauge check.

12. Report Writing Rules

When the assistant generates an inspection report, the root-cause section must separate observed evidence, retrieved evidence, possible cause, and recommended checks.

Required report structure

Report field	Required content
Observed anomaly	Use the vision output: product category, anomaly status, location, confidence, and severity.
Retrieved evidence	Mention which SOP or troubleshooting section was retrieved.
Possible root cause	List possible cause families. Do not present a single guaranteed cause unless confirmed by external data.
Recommended checks	List practical inspection checkpoints such as machine settings, tooling, material, handling, or cleaning.
Immediate action	Accept, rework, reject, quarantine, retake image, human review, or escalate process.
Limitation statement	Mention if process data was not available or image quality was low.

Example safe report text

The image shows a suspected crack on the bottle rim with high visual confidence. Based on the Bottle Root-Cause Troubleshooting section, rim cracks may be associated with molding/forming parameters, cooling variation, material weakness, or handling impact. Because the rim can affect sealing quality, the item should be rejected and the batch should be checked for repeated rim defects. Recommended checkpoints include mold pressure, cooling records, conveyor impact points, and packaging handling. The root cause is not confirmed from image data alone and should be verified with process records.

13. Final Notes for the Project Knowledge Base

This guide should be stored in the RAG document folder together with the QA inspection manual, defect severity guide, and operator action SOP. The best RAG answer should retrieve at least one category-specific troubleshooting section and one action or severity section before producing the final recommendation.

Recommended file name: Root_Cause_Troubleshooting_Guide.pdf

Recommended folder: rag_documents/

Recommended document type metadata: troubleshooting_guide

End of document